

Log-Free Hyperelliptic Specialization and Class Groups of Quadratic Fields

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Abstract

We prove that there are $\gg X^{1/3}$ imaginary quadratic fields K with discriminant $|\text{Disc } K| \leq X$ and an ideal class group of 5-rank at least 2. This removes the factor $(\log X)^{-2}$ from the lower bound of Bartz, Levin, and Thamminana. We obtain this by applying a stronger squarefree-value theorem of Stewart and Top to refine the quantitative specialization result of Kulkarni and Levin, removing the logarithmic loss when every Galois orbit in the branch locus of the degree-two map has size at most 6.

Keywords: quadratic fields; ideal class groups; hyperelliptic curves; power-free values; Hilbert irreducibility.

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1 Introduction

Given an integer $n > 1$, one may ask not only whether n divides the class number of infinitely many quadratic fields, but more generally how often the ideal class group contains several independent elements of n -torsion. For a finite abelian group A , we define its n -rank by

$$\text{rk}_n A = \max\{r \geq 0 : (\mathbf{Z}/n\mathbf{Z})^r \hookrightarrow A\}.$$

For $X \geq 1$, let $N^-(n^r; X)$ and $N^+(n^r; X)$ denote the numbers of imaginary and real quadratic fields K , respectively, counted up to $\mathbf{Q}$ -isomorphism, satisfying

$$|\text{Disc } K| \leq X, \quad \text{rk}_n \text{Cl}(K) \geq r.$$

One approach to these counting problems is through specialization on algebraic curves. Rational torsion on the Jacobian of a curve can, under suitable specialization, give rise to torsion in the ideal class groups of number fields. In the quadratic setting, Kulkarni and Levin [3, Theorem 1.3] obtained quantitative lower bounds from hyperelliptic curves with a rational Weierstrass point. In the setting of their theorem, if $C/\mathbf{Q}$ has genus $g \geq 1$ and

$$\rho = \text{rk}_n \text{Jac}(C)(\mathbf{Q})_{\text{tors}},$$

then

$$N^-(n^\rho; X) \gg_{C,n} \frac{X^{1/(g+1)}}{(\log X)^2},$$

and, when $\rho \geq 1$,

$$N^+(n^{\rho-1}; X) \gg_{C,n} \frac{X^{1/(g+1)}}{(\log X)^2}.$$

A recent application of this method is due to Bartz, Levin, and Thamminana [1]. They constructed a genus-two curve with a rational Weierstrass point whose rational Jacobian torsion has 5-rank at least two. Applying the quantitative specialization result of Kulkarni and Levin, they obtained

$$N^-(5^2; X) \gg \frac{X^{1/3}}{(\log X)^2}.$$

This improved an earlier unconditional lower bound of order $X^{1/4}$ due to Byeon [2].

The factor $(\log X)^{-2}$ in the Kulkarni–Levin argument is inherited from the Stewart–Top estimate used to count binary-form values modulo squares. When every irreducible factor of the relevant binary form has degree at most six, the stronger squarefree-value argument of Stewart and Top [5, Theorem 1] applies. We use this stronger estimate within the specialization framework of Kulkarni and Levin to remove the logarithmic loss.

The factor-degree condition has a natural interpretation in terms of the branch points of the double cover. Let $C/\mathbf{Q}$ be a smooth projective curve of genus $g \geq 1$, equipped with a degree-two morphism $\pi : C \rightarrow \mathbb{P}^1$ defined over $\mathbf{Q}$ and having a rational ramification point. We define

$$\mu(C, \pi) = \max_{\beta} \#(\text{Gal}(\overline{\mathbf{Q}}/\mathbf{Q}) \cdot \beta),$$

where β ranges over the geometric branch points of π .

Our main result is the following.

Theorem 1.1. *Let $C/\mathbf{Q}$ be a smooth projective curve of genus $g \geq 1$, equipped with a degree-two morphism*

$$\pi : C \rightarrow \mathbb{P}^1$$

defined over $\mathbf{Q}$ and having a rational ramification point. Suppose that

$$\mu(C, \pi) \leq 6.$$

Let $n > 1$, and put

$$\rho = \text{rk}_n \text{Jac}(C)(\mathbf{Q})_{\text{tors}}.$$

Then

$$N^-(n^{\rho}; X) \gg_{C,n} X^{1/(g+1)}.$$

If $\rho \geq 1$, then

$$N^+(n^{\rho-1}; X) \gg_{C,n} X^{1/(g+1)}.$$

By the Riemann–Hurwitz formula, the branch locus of π consists of $2g + 2$ geometric points. Hence

$$\mu(C, \pi) \leq 2g + 2,$$

and the hypothesis $\mu(C, \pi) \leq 6$ is automatic for $g \leq 2$. In particular, applying Theorem 1.1 to the genus-two curve of Bartz, Levin, and Thamminana gives the following.

Corollary 1.2. *There are $\gg X^{1/3}$ imaginary quadratic fields K satisfying*

$$|\text{Disc } K| \leq X, \quad \text{rk}_5 \text{Cl}(K) \geq 2.$$

2 Branch forms

Choose a rational branch point as the point at infinity and write

$$C : y^2 = f(x), \quad \deg f = 2g + 1,$$

with $f \in \mathbf{Q}[x]$ squarefree. Set

$$\mathcal{B}_C(X, Y) = Y^{2g+2} f(X/Y).$$

Lemma 2.1. *The multiset of Galois-orbit sizes in the branch divisor of C is invariant under a change of hyperelliptic coordinate by an element of $\mathrm{PGL}_2(\mathbf{Q})$. In particular, $\mu(C)$ is independent of the chosen model and of the rational branch point placed at infinity.*

Proof. A rational fractional-linear transformation induces a $\mathrm{Gal}(\overline{\mathbf{Q}}/\mathbf{Q})$ -equivariant bijection of $\mathbf{P}^1(\mathbf{Q})$ and hence preserves the orbit sizes in the branch divisor. For a reduced binary form over $\mathbf{Q}$, these sizes are the degrees of its irreducible factors.

If $g \geq 2$, uniqueness of the hyperelliptic involution implies that any two hyperelliptic coordinates differ by an element of $\mathrm{PGL}_2(\mathbf{Q})$. If $g = 1$, choosing a rational branch point as the origin identifies the ramification set with $E[2]$; changing the origin by a rational branch point preserves its Galois action. We henceforth write $\mu(C)$ for this invariant. $\square$

Lemma 2.2. *Let $f \in \mathbf{Q}[x]$ be nonzero and $c \in \mathbf{Q}$. For every $N_0 > 0$ there is an integer $N > N_0$ such that*

$$f(c + 1/N)f(c - 1/N) \neq 0.$$

Consequently, the parameter N in [3, Lemma 3.1] may be chosen so that the transformed branch form has nonzero endpoint coefficients.

Proof. For each root β of f , either equation $c \pm 1/N = \beta$ excludes at most one value of N . Thus only finitely many positive integers are excluded, whereas N may be chosen arbitrarily large in the proof of [3, Lemma 3.1]. $\square$

Put

$$\psi_N(s) = \frac{1 - Ns}{1 + Ns}, \quad \tau_N(q) = \psi_N^{-1}(q) = \frac{1 - q}{N(1 + q)}. \quad (1)$$

For $q = X/Y$, set

$$L_1 = Y - X, \quad L_2 = N(X + Y),$$

and choose $\lambda \in \mathbf{Q}^\times$ so that

$$F(X, Y) = \lambda^2 L_2^{2g+2} f\left(c + \frac{L_1}{L_2}\right) = \lambda^2 \mathcal{B}_C(cL_2 + L_1, L_2) \in \mathbf{Z}[X, Y]. \quad (2)$$

Since $2g + 2$ is even,

$$\mathbf{Q}\left(\sqrt{F(a, b)}\right) = \mathbf{Q}\left(\sqrt{f(c + \tau_N(a/b))}\right) \quad (3)$$

whenever the expressions are defined. By Lemmas 2.1 and 2.2, the largest degree of an irreducible factor of F is $\mu(C)$ and

$$F(1, 0)F(0, 1) \neq 0.$$

3 Squarefree values

Proposition 3.1. *Let $F \in \mathbf{Z}[X, Y]$ be a binary form of degree $r \geq 3$ with nonzero discriminant and*

$$F(1, 0)F(0, 1) \neq 0.$$

Suppose that every irreducible factor of F over $\mathbf{Q}$ has degree at most 6. Fix integers A, B, M with $M \geq 1$, and let w be the largest positive integer such that

$$w^2 \mid F(a, b)$$

whenever

$$a \equiv A \pmod{M}, \quad b \equiv B \pmod{M}.$$

Then, for all sufficiently large H , there are $\gg H^2$ distinct nonzero squarefree integers t for which

$$F(a, b) = tw^2 \tag{4}$$

has a solution satisfying

$$1 \leq a, b \leq H, \quad a \equiv A \pmod{M}, \quad b \equiv B \pmod{M}.$$

The implied constant depends only on F, A, B, M .

Proof. This is the case $k = 2$ of the argument proving [5, Theorem 1]. In the notation of Stewart–Top, the congruence conditions

$$a \equiv A \pmod{M}, \quad b \equiv B \pmod{M}$$

are already built into the counting problem. For $k = 2$, their argument applies when every irreducible factor of F has degree at most 6 and yields $\gg H^2$ admissible pairs in the prescribed box for which $F(a, b)/w^2$ is squarefree.

For these pairs, $\gcd(a, b) \mid w$. After fixing the common divisor and passing to primitive pairs, the representation estimates in [5, pp. 952–953], based on [4, Corollary 1], show that these pairs yield $\gg H^2$ distinct squarefree represented values. $\square$

4 Proof of the main theorem

Proof of Theorem 1.1. Let f be as above, put $d = 2g + 1$, and let S be the finite set of primes used in the proof of [3, Theorem 1.3]. Set

$$M_0 = \prod_{p \in S} p,$$

with $M_0 = 1$ if $S = \emptyset$.

Choose a sufficiently large positive integer u , coprime to M_0 , and take

$$c = \pm \frac{u}{M_0},$$

with the sign chosen so that $f(c)$ has the required sign for the real or imaginary case. Since f has odd degree, this is possible for all sufficiently large u . Moreover, for $p \in S$,

$$v_p(c) = -1, \quad v_p(f(x)) = -d$$

on a sufficiently small p -adic neighborhood of c . Since d is odd, every $p \in S$ ramifies in $\mathbf{Q}(\sqrt{f(x)})$.

Apply [3, Theorem 4.1] to $\phi = x - c$. There is a fixed thin set $\Omega \subset \mathbf{Q}$ outside which the class-group specialization estimate holds. Using the prescribed finite and archimedean local conditions, the construction in [3, Lemma 3.1], together with Lemma 2.2, gives a parameter N , a modulus M , and a positive residue class, which we take to be

$$a \equiv b \equiv 1 \pmod{M}.$$

The associated form F in (2) has degree $D = 2g + 2$, nonzero discriminant, nonzero endpoint coefficients, and largest irreducible-factor degree $\mu(C) \leq 6$. Let w be the maximal fixed square divisor of its values on the chosen residue class. By Proposition 3.1, for all sufficiently large H there are $\gg H^2$ distinct signed squarefree integers t with

$$F(a_t, b_t) = tw^2, \quad 1 \leq a_t, b_t \leq H, \quad a_t \equiv b_t \equiv 1 \pmod{M}.$$

Put

$$q_t = \frac{a_t}{b_t}, \quad \alpha_t = \tau_N(q_t).$$

Distinct t give distinct α_t , and the α_t have height $O(H)$. By [3, Lemma 3.2], only $O(H)$ of them lie in Ω , so $\gg H^2$ values remain.

For each surviving t , choose

$$P_t = \left(c + \alpha_t, \sqrt{f(c + \alpha_t)} \right) \in C(\overline{\mathbf{Q}}), \quad K_t = \mathbf{Q}(P_t).$$

Then $\phi(P_t) = \alpha_t \notin \Omega$, so [3, Theorem 4.1] gives $[K_t : \mathbf{Q}] = 2$ and the class-group specialization estimate. By (3),

$$K_t = \mathbf{Q}\left(\sqrt{f(c + \alpha_t)}\right) = \mathbf{Q}(\sqrt{t}).$$

Thus distinct surviving signed squarefree values of t give distinct quadratic fields.

Every $p \in S$ is ramified in K_t . Dirichlet's unit theorem gives

$$\mathrm{rk} \mathcal{O}_{K_t, S}^\times = \begin{cases} \#S, & K_t \text{ imaginary,} \\ \#S + 1, & K_t \text{ real.} \end{cases}$$

By [3, Theorem 4.1],

$$\mathrm{rk}_n \mathrm{Cl}(K_t) \geq \rho + \#S - \mathrm{rk} \mathcal{O}_{K_t, S}^\times,$$

and therefore

$$\mathrm{rk}_n \mathrm{Cl}(K_t) \geq \begin{cases} \rho, & K_t \text{ imaginary,} \\ \rho - 1, & K_t \text{ real.} \end{cases}$$

The archimedean condition ensures that $f(c + \alpha_t)$, and hence t , has the required sign, since the two radicands in (3) differ by a rational square. Finally,

$$|t| = \frac{|F(a_t, b_t)|}{w^2} \ll H^D,$$

so

$$|\text{Disc } K_t| \ll H^D.$$

Taking H to be a sufficiently small fixed multiple of $X^{1/D}$ gives

$$H^2 \asymp X^{2/D} = X^{1/(g+1)},$$

which proves the theorem. □

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